USC Dornsife
Dana and David Dornsife
College of Letters, Arts and Sciences
Spatial Sciences Institute

SSCI 420 (35668)
Human Security and Geospatial Intelligence Practicum

Syllabus

Units: 4

Term Day Time: Spring 2025, Wednesday, 2:00-5:50pm

Location: DMC 200

Instructor: Diana Ter-Ghazaryan, PhD, GISP
Office: AHF B55K
Regular Office Hours: Mondays 11:00am-1:00pm
Also available most days and times by appointment via email.
Contact Info: terghaza@usc.edu, 213-821-1190

Library Help: Andy Rutkowski
Office: LIPA B40-A
Office Hours: TBD
Contact Info: arutkows@usc.edu

IT Help: Spatial Support
Contact Info: spatial_support@usc.edu.

Course Scope and Purpose

The Human Security and Geospatial Intelligence Practicum is designed to provide students a hands-on problem-solving opportunity that requires them to employ knowledge and GEOINT skills to provide decision-makers an informed recommendation involving a variety of human security settings. Threats to human security come in many forms – military operations, terrorist attacks, genocide, political violence, natural disasters, humanitarian crises, environmental risks, public health issues and food / resource accessibility challenges, among others – and this class leverages a variety of exposures to geospatial solutions for the intelligence community and intelligence products that support national security, disaster response, and humanitarian relief efforts.

The outline that follows provides the structure and rules that should be necessary to embark on this ambitious 15-week project. For a successful project, it is important that:

- Students operate as a professional team with the support and guidance of the faculty instructor;
- All parties develop a high level of trust and spirit of collaboration;
  students and faculty are responsive to clients while conducting independent research;
- Communication between parties is professional and frequent; and students feel comfortable to make decisions, make mistakes, and learn from the experience.

Learning Outcomes

When you have completed this course, you will be able to:

- Reinforce the understanding of geospatial intelligence applications of Human Security and GEOINT as related to government, industry, and academe.
- Design and implement strategies for capturing or sourcing geospatial intelligence data and any accompanying metadata.
- Critically evaluate the potential impacts of data quality on spatial analysis and decision making to the intelligence community.
- Apply critical thinking, collaboration, and communication skills.
- Synthesize learning by preparing and presenting a project report on Geospatial Intelligence, tailored to a specific, real-world human security application.

Students may vary in their competency levels on these abilities. You can expect to acquire these abilities only if you honor all course policies, attend classes regularly, complete all assigned work in good faith and on time, and meet all other course expectations of you as a student.

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Prerequisite(s): SSCI 202: Principles of Geospatial Intelligence
Co-requisite(s): None
Concurrent Enrollment: None
Recommended Preparation: SSCI 301: Maps and Spatial Reasoning

Class Conduct

Harassment, sexual misconduct, interpersonal violence, and stalking are not tolerated by the university. All faculty and most staff are considered Responsible Employees by the university and must forward all information they receive about these types of situations to the Civil Rights Coordinator. The Civil Rights Coordinator is responsible for assisting students with supportive accommodations, including academic accommodations, as well as investigating these incidents if the reporting student wants an investigation. The Office of Civil Rights is also responsible for coordinating supportive measures for transgender and nonbinary students such as faculty notifications, and more. If you need supportive accommodations, you may contact the OCRC directly (ocrc@usc.edu or 213-740-5086) without sharing any personal information with me. If you would like to speak with a confidential counselor, Relationship and Sexual Violence Prevention Services (RSVP) provides 24/7 confidential support for students (213-740-9355 (WELL); press 0 after hours)

USC Culture Journey is an initiative aimed at fostering a more **inclusive, values-based, and integrity-driven culture** across the university. The components of the Culture Journey include: community engagement, core values, strategic cultural goals, ongoing assessment, and leadership commitment. Use the following link to see full details on the Culture Journey: https://culturejourney.usc.edu/

Course Structure

Academic Units and Grading

Students will work in groups but will be graded separately. It is *not* required that all students in a project receive the same grade.

At the end of the semester, students are required to complete self-evaluations and peer evaluations and submit them to their instructor by the last day of classes. The faculty instructor will consider these evaluations in assigning grades.

Grades will not be assigned until the group has submitted and the faculty instructor has approved the final report.

Student Time Commitment

Students should expect to devote, on average, 10 hours a week to the Group Project. This time includes class meeting time and field trips as appropriate for each project. Progress on Group Projects should be evenly allocated over the semester to avoid excess workload during the last few weeks.

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Required Work Products

The required work products for each group include:

- A literature review and project proposal
- Proposal presentation (including data report)
- Draft report
- Final report, including an executive summary
- Presentation of the final report
- Peer evaluations.

Authorship/Ownership

Each member of the group, along with the faculty instructor for that group, is an equal owner of the data collected during the project. That means that each member of the group can use the data even after the project has ended. Papers, posters, presentations, and other derivative products that arise from the data collected during the project must acknowledge every member of the group in a manner that is appropriate to the group's contribution to the derivative product.

Data Distribution

Datasets and processed datasets that may have been obtained/derived during your work may be licensed or copyrighted or confidential. You should *not* make them available to third parties without authorization from your faculty instructor, Group Project advisor, *and* the original source of data. Also, data should not be placed online unless your faculty instructor, Group Project advisor, and the owner of data have properly approved doing so.

Use of Human Subjects

Faculty and students that engage in research involving human subjects must first obtain prior approval from the Institutional Review Board (IRB). "Human Subject" means a living individual about whom an investigator (whether professional or student) conducting research obtains (i) data through intervention or interaction with the individual or (ii) identifiable private information. This means that even if you are just going to be conducting a survey, you MUST obtain IRB approval in advance. Approval is required no matter with whom you will be interacting — even your friends or family!

Technological and Communication Requirements

ArcGIS is provided online via the GIST Server; hence, you do not need to install it on your own computer. Instead, every student must have the following technology requirements:

- A computer with a fast Internet connection.
- A functional webcam and a microphone for use whenever a presentation or meeting is scheduled.
- An up-to-date web browser to access the Server

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If a student does not have access to any of these, please speak with the instructor at the start of the semester. Also, see the USC ITS Student Toolkit here: https://keepteaching.usc.edu/students/student-toolkit/

A limited number of computers with all the necessary software is available in the SSI Suite (AHF B55) during regular business hours, Monday through Friday 9 am to 5 pm. These computers are available to any student in an SSCI or GSEC course, first-come first-served, and can be used as a resource if you experience difficulties in accessing the SSI server or using the GIS software on your personal computer.

Brightspace – This course will utilize the Brightspace learning management system which allows students to access course content, upload assignments, participate in discussion forms, among other learning experiences. The Brightspace platform provides flexibility in the learning experience where students can participate in the course residentially or remotely, synchronously (meeting together at the same time) or asynchronously (accessing videos and course content outside of class).

SSI Server and Tech Support – This course utilizes the SSI Server which is a virtual desktop giving access to many different professional software. If you are unable to connect to the server or experience any type of technical issues, send an email using your USC account to SSI Tech Support at spatial_support@usc.edu, making sure to copy (cc) me on the email.

Communications – All assignments given and all materials to be handed in will be submitted via Brightspace The instructor will also create and monitor discussion forums through which students can discuss issues and assignments as needed. Students should read all email sent from Brightspace or from course instructor(s) as soon as possible. Also, students who do not regularly use their USC email accounts should double-check to be sure that mail sent from both the Brightspace accounts and the instructor's account (noted above) to your USC account is forwarded to an address used regularly and does not go into junk mail. The instructor will endeavor to respond to all email within 24 hours of receipt, aiming for no more than 72 hours delay. In the rare case that an instructor is off-line for an extended period of time, an announcement will be posted to the class Brightspace site. Due to the synchronous and asynchronous nature of this course, it is each student's responsibility to stay informed and connected with others in our course. In addition to email, students are expected to login to Brightspace regularly to check for announcements.

Required Readings and Supplementary Materials

Textbooks – There are two required texts for this course. Some (of the optional texts) are available online and some are available from the USC Bookstore or online outlets such as Amazon. It is best to acquire these materials quickly since you will need them from the opening day of class.

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- Clark, R. 2020. *Geospatial Intelligence – Origins and Evolution.* Georgetown Washington, DC: University Press (find at: http://press.georgetown.edu/book/georgetown/geospatial-intelligence).
- NGA (National Geospatial-Intelligence Agency, Office of Geospatial-Intelligence Management). 2018. *National System for Geospatial Intelligence: Geospatial Intelligence (GEOINT) Basic Doctrine.* Washington, DC: National Geospatial-Intelligence Agency Publication No. 1-0 (FREE download at https://www.nga.mil/ProductsServices/Pages/GEOINT-Basic-Doctrine-Publication.aspx)

The textbooks will be supplemented with course notes, videos, and a mixture of planned readings from academic journals, professional reports, and authoritative websites. The list below is a sample of some of the readings that may be used. Additional readings not in the list below may also be provided.

- Esri. 2012. ArcGIS for Emergency Management, An Esri White Paper – May 2012. Redlands, CA: Esri Press.
- Esri. 2014. GIS Platform for National Security, An Esri White Paper – July 2014, Redlands, CA: Esri Press.
- Esri. 2007. GIS Supporting the Homeland Security Mission, An Esri White Paper May 2007, Redlands, CA: Esri Press.
- Evans, Howard, James Lange, and James Schmitz. 2015. *The Phenomenology of Intelligence-focused Remote Sensing (selected readings).* Beavercreek, OH, Riverside Research.
- Gillespie, T., Chu, J., Frankenberg, E., Thomas, D. 2007. "Assessment and prediction of natural hazards from satellite imagery." *Progress in Physical Geography,* 31(5): 459-470.
- Lowenthal, M.M. (2016) *Intelligence: From Secrets to Policy* (7th Edition). Washington, DC, CQ Press.
- NRC (National Research Council) (2007) *Successful Response Starts with a Map: Improving Geospatial Support for Disaster Management.* Washington, DC, National Academies Press.
- United States Geospatial Intelligence Foundation (USGIF 1). 2017. *Trajectory Magazine –Public Safety Edition.* Reston, VA: USGIF Press.
- United States Geospatial Intelligence Foundation (USGIF 2). 2018. *Building Resilient Communities Through Geospatial Intelligence.* Reston, VA: USGIF Press.

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- United States Geospatial Intelligence Foundation (USGIF 3). 2019 (and previous years). *The State and Future of GEOINT*. USGIF, Herndon, VA, USGIF Press.

Grading Breakdown

Careful planning and a serious, consistent commitment will be required for you to successfully navigate the various deliverables in this and other courses. This table summarizes the SSCI 420 course assignments and percentage weight of the final grade:

Assignments	**Points**
Literature Review	15
Project Proposal	20
Data Report	10
Proposal Presentation	5
Draft Report	10
Final Presentation/Client Presentation	10
Final Report	20
Self and Peer Evaluation	10
Total:	**100**

Grading Scale

Assignments in this and other SSCI courses, are graded on the letter grade scale where A is exemplary, B is very good, C is satisfactory, D is unsatisfactory, and F needs improvement. Final grades use the same letter grade scale with C being the minimum passing grade for credit at the graduate level. The grading scale follows:

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A	> 93 points		B-	80-82 points		D+	67-69 points
A-	90-92 points		C+	77-79 points		D	63-66 points
B+	87-89 points		C	73-76 points		D-	60-62 points
B	83-86 points		C-	70-72 points		F	<60 points

Assignment Submission Policy

Assignments will be submitted for grading via Brightspace using the due dates specified in the Course Schedule below.

SSI Policy on the Creation of Original Work and Use of Generative AI

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. Students may not have another person or entity complete any substantive portion of an assignment or reuse work prepare for courses without obtaining written permission from the instructor(s). Developing strong competencies in research, writing, and the technical execution of geospatial technologies are foundational to SSI academic programs that are designed to prepare you for success in the workplace. Therefore, using generative AI tools – unless explicitly specified otherwise – is strictly prohibited in this course, will be identified as plagiarism, and will be reported to the Office of Academic Integrity.

Grading Timeline

My goal is to provide grading and feedback on each course assignment in time for you to take my feedback into consideration as the course progresses. Generally, this means that you can expect feedback within 1 week after a given assignment's due date.

Learning Experience Evaluations

Please note Learning Experience Evaluations for the course take place at the end of the semester and are facilitated by the University. These evaluations provide an important review of student experiences in the course.

Course Content Distribution and Synchronous Session Recordings Policies

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.

Recording a university class without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an Office of Accessibility Services

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(OSAS) accommodation. Recording can inhibit free discussion in the future, and thus infringe on the academic freedom of other students as well as the instructor. (Living our Unifying Values: The USC Student Handbook, page 13).

Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposed other than individual or group study is prohibited. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which has been distributed to students or in any way has been displayed for use in relationship to the class, whether obtained in class, via email, on the internet, or via any other media. (Living our Unifying Values: The USC Student Handbook, page 13).

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Course Schedule: A Weekly Breakdown

	Topic	Assignments and Readings	Deliverables/Due Dates
Week 1 1/14	**Introduction** What is Human Security and Geospatial Intelligence?	Reading: NGA Basic Doctrine; NRC Successful Response; "What is Geospatial Intelligence?" Assignment: none	No deliverables
Week 2 1/21 *Monday, 1/19 is university holiday	**Semester Outlook** Human Security and Geospatial Intelligence as Collaborative Decision-making Framework Project Definition and Details	Reading: USGIF EBK (2019) Assignment: Literature Review	No deliverables
Week 3 1/28	**Client Introduction/Visit** Meet client and learn background and context for the semester-long project. Group discussion on various aspects of organizing, managing, and executing a research project.	Reading: Assignment:	No deliverables
Week 4 2/04	**Project Overview** Discuss project overview, brainstorm required skills, and project approach(es).	Reading: Assignment: Project Proposal	Due: Literature Review
Week 5 2/11	**Project Update** Provide reports on literature review and projects; citation management; and brainstorm screening criteria.	Reading: Assignment:	No deliverables
Week 6 2/18* *Monday, 2/16 is university holiday	**Project Update** Continued	Reading: Assignment:	Due: TBD

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	Topic	Assignments and Readings	Deliverables/Due Dates
Week 7 2/25	**Project Proposal Presentations** Present project proposal and discuss peer proposals.	Reading: Assignment:	No deliverables
Week 8 3/04	**Project Constraints** Discuss project constraints, literature reviews, assign stakeholder advocates, define stakeholder data topics, and stakeholder participation.	Reading: Assignment:	Due: TBD
Week 9 3/11	**Client Update** Meet with client to discuss progress on design challenge project. Discuss constraint and opportunity models	Reading: Assignment:	No deliverables
3/18 *3/16-3/20 is Spring Recess	None	None	No deliverables
Week 10 3/25	**Project Update** Applying Human Security and Geospatial Intelligence as Collaborative Decision-making Framework	Reading: Assignment:	Due: Intermediate Report
Week 11 4/01	**Project Update** Continued	Reading: Assignment:	No deliverables
Week 12 4/08	**Project Work Session** Students work in groups on their research project.	Reading: Assignment:	No deliverables

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	Topic	Assignments and Readings	Deliverables/Due Dates
Week 13 4/15	**Presentation and Discussion of Draft Reports** Students present and discuss their draft reports.	Reading: Assignment:	Due: Rough Draft Final Report
Week 14 4/22	**Project Work Session** Students work in groups on their research project.	Reading: Assignment:	Due: Peer and Self-Reviews due
Week 15 4/29 Friday, 5/1 is the last day of class	**Final Presentation/Client Presentation**		Due: Final Presentation
Final Exams 5/6-5/13	**Final Project Report**		Due: Final Report

Statement on Academic Conduct and Support Systems

Academic Integrity

The University of Southern California is a learning community committed to developing successful scholars and researchers dedicated to the pursuit of knowledge and the dissemination of ideas. Academic misconduct, which includes any act of dishonesty in the production or submission of academic work, comprises the integrity of the person who commits the act and can impugn the perceived integrity of the entire university community. It stands in opposition to the university's mission to research, educate, and contribute productively to our community and the world.

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. You may not submit work written by others (including AI generated) or "recycle" work prepared for other courses without obtaining written permission from the instructor(s).

Other violations of academic integrity include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), collusion, knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

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The impact of academic dishonesty is far-reaching and is considered a serious offense against the university. All incidences of academic misconduct will be reported to the Office of Academic Integrity and could result in outcomes such as failure on the assignment, failure in the course, suspension, or even expulsion from the university.

For more information about academic integrity see the student handbook or the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct.

Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

SSI Policy on the Creation of Original Work and Use of Generative AI

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. Students may not have another person or entity complete any substantive portion of an assignment or reuse work prepare for courses without obtaining written permission from the instructor(s). Developing strong competencies in research, writing, and the technical execution of geospatial technologies are foundational to SSI academic programs that are designed to prepare you for success in the workplace. Therefore, using generative AI tools – unless explicitly specified otherwise – is strictly prohibited in this course, will be identified as plagiarism, and will be reported to the Office of Academic Integrity.

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

988 Suicide and Crisis Lifeline - 988 for both calls and text messages – 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for

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people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

CARE-SC - (213) 740-9355(WELL) – 24/7 on call

Free and confidential therapy services, workshops, and training for situations related to gender- and power-based harm (including sexual assault, intimate partner violence, and stalking).

Office for Civil Rights Compliance - (213) 740-5086

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 740-0411

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

USC Culture Journey - (213) 740-8258

At USC our six unifying values support our University's mission, and guide our behaviors, the way we engage with ourselves and our community, to create our Trojan culture.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-1200 – 24/7 on call

Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.

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